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Clinically feasible dual-task detects residual post-concussion deficits after return-to-play

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ABSTRACT

Dual-task (DT) has been shown to detect post-concussion deficits even after traditional measures returned to normal. However, previous studies were conducted in laboratory settings that were not feasible in the clinical setting. Prospective Cohort Study. Two groups (concussion and control groups) completed two testing sessions. The concussion group was tested at 7–10 days post-concussion (subacute) and after Return-to-Play (RTP). The control group was tested with similar time windows. Participants completed 2 single task (ST): Expanded Timed Get Up and Go (ETGUG) as a physical task and Auditory Pure Switch Task (APST) as a cognitive task. After completing STs, participants completed DT performing ETGUG and APST simultaneously. A mixed method ANOVA was conducted to analyze the effects of concussion on ST and DT performance. Another mixed method ANOVA was conducted to examine the effect of concussion recovery time. The concussion group took significantly longer to complete ST and DT than controls at the subacute phase ($p = 0.018$, $p = 0.014$, respectively). At RTP timepoint, the concussion group took significantly longer to complete ST and DT than the controls (p=0.007, $p < 0.001$, respectively). The deficits were more clearly shown for DT. Clinically feasible DT detected subtle and residual post-concussion deficits after athletes were cleared to RTP.

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Concussion; divided attention; dual-task; single-task

Introduction

Current concussion assessment tools commonly utilized in clinical settings are designed to assess the individual's ability to perform one task at a time, known as a single task (ST). These STs may not fully challenge the complicated neurocognitive functions that are required in daily and sports activities. While the Return-to-Learn and Return-to-Play (RTP) protocols are well established and universally accepted, about half of the individuals with concussion reported recurrent symptoms or worsening of symptoms after returning to school and sports activities (Carson et al. 2014). School and sports activities often require performing multiple tasks at the same time, switching between tasks, and keeping attention despite distraction (Carlson et al., 2013). These skills are a function of one of the highest cognitive processes referred to as executive function (Carlson et al., 2013). A specific example of executive function is the ability to perform more than one task simultaneously, called divided attention (Sohlberg & Mateer, 1987). Deficits in divided attention are commonly reported by patients suffering from concussions (Cicerone, 1996); however, the current concussion assessment protocol utilized in clinical settings is not designed to assess divided attention.

Dual-task (DT) is designed to challenge divided attention and it is the closest tool currently available to replicate the type of cognitive processing used in school and sports activities (Bowen et al., 2001). The DT performance in individuals with concussion has been investigated in the laboratory setting (Fait et al., 2009; Lee et al., 2013; Parker et al., 2006). These studies reported that the DT performance of individuals with concussion was significantly altered compared to that of the non-concussed individuals. Furthermore, DT has been shown to detect post-concussion deficits even after traditional measures such as balance scores, neurocognitive scores, and symptom scales have returned to normal (Fait et al., 2009) indicating that recovery of divided attention may take longer than that of simpler cognitive and physical functions. Howell et al. investigated DT's ability to detect post-concussion deficits prospectively up to 2 months post-concussion (Howell et al., 2013). In this study, the Stroop test was used as a cognitive task and self-selected speed walking was used as a physical task. They found that the medial-lateral displacement and peak medial-lateral velocity of the center of mass during DT were greater in the athletes with concussion than in the control participants, even at 2-months post-concussion. These residual deficits may explain the increase in the risk of musculoskeletal injuries up to a year following concussion (Lynall et al., 2015, 2017). The activities of daily living and sports require simultaneous multisensory information processing, which is not evaluated with standardized concussion assessments. Therefore, DT testing may be a useful tool to assess the readiness for return to daily and sports activities.

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Previous studies investigating DT assessments were conducted in laboratory settings that were not readily available in most clinical settings. One of the goals of having DT as an adjunct concussion assessment is to be clinically feasible, less time-consuming, and sensitive to identifying post-concussion deficits more precisely. Only a few studies investigated whether DT can be used in traditional clinical settings (Hashida et al., 2021; Howell et al., 2019). Tamura et al. compared the reliability of different DT combinations in healthy college students in an attempt to find the most reliable cognitive task to compose a clinically feasible DT (Tamura et al., 2018). The result from the study showed that Expanded Timed Get Up and Go (ETGUG) combined with Auditory Pure Switch Task (APST) was the most clinically reliable DT combination. APST is a relatively simple task that requires a participant to answer if a given number is odd or even, and has been reported to have high reliability with no practice effect when used under DT conditions (Banks, 2006; Okumura et al., 2013). Additionally, the APST requires less language skills than other complex cognitive tasks. The study was conducted on non-concussed college students, and the efficacy of DT using ETGUG and APST in concussion assessment remains unknown.

Based on the current evidence, it is likely that some portion of athletes with concussion are cleared to RTS while experiencing 'undetected' post-concussion deficits, as it has been reported that athletes with concussion displayed post-concussion deficits under DT condition even at 2-months post-concussion (Cicerone, 1996). Therefore, the purpose of this study was to compare clinically feasible ST and DT ETGUG outcomes between concussion and matched control groups at the subacute phase and after completion of the RTP protocol. It was hypothesized that DT performance in the concussion group would be significantly worse than that of the matched control group at both subacute and after completion of RTP, which would not be detected by using ST.

Methods

Study Design

This study utilized a 2 × 2 (between and within subjects) mixed design. Two groups (concussion and control groups) completed two separate testing sessions where both ST and DT were performed. The concussion group was tested at 7–10 days post-concussion (subacute) and after the RTP protocol. RTP was defined when the athlete completed a standardized graduated RTP protocol monitored by school's athletic trainers. The control group was tested with similar time windows. Data collection occurred in quiet indoor facilities with minimal distractions at either the local high school's athletic training rooms or laboratories. A single researcher administered 2 trials of ST, followed by DT for each participant.

Participants

Athletes with concussion were recruited through local high schools and universities, where they were diagnosed with concussion and referred by sports medicine physicians. Non-concussed controls were recruited from the same schools.

Table 1. Demographic data for concussion and control group.

	Concussion	Control	*p*-value
Male (n)	10	10	
Female (n)	10	10	
Age (y)	16.94 ± 3.09	16.95 ± 2.59	1.000
Height (cm)	174.32 ± 14.26	175.12 ± 15.77	0.912
Body Mass (kg)	68.17 ± 14.10	72.22 ± 22.43	0.652

Previous research showed that females and males have different gait patterns (Howell et al., 2017) and DT performance differed between athletes who participated in contact and non-contact sports (Horak, 2006). Thus, each athlete with concussion was matched with non-concussed control by age, sex, and sport. The inclusion criteria for the concussion group were: between ages 12 and 49 years old, diagnosed with a concussion, and can walk a 20-meter distance. Also, participants with a history of lower extremity injuries that cause abnormal gait, active neurological disorder, history of cognitive deficiency, existing psychiatric condition, or other conditions that may affect the result of this study, were excluded.

A total of 41 individuals agreed to participate in the study, however, one athlete with a concussion was not available for follow-ups. A total of 20 athletes with concussion (age: 16.94 ± 3.09 years, 50% female) and 20 non-concussed controls (age: 16.95 ± 2.59 years, 50% female) completed the entire data collection. The demographic characteristics of all participants in each group are presented in Table 1. No significant differences between groups were found for demographic information (all $p \geq 0.05$).

Procedure

Upon arrival at the data collection site, participants completed an informed consent/assent form approved by the University's Human Studies Program Institutional Review Board along with a background questionnaire. All participants completed a symptoms assessment using a Graded Symptom Checklist (GSC) which has been used to assess the number and severity of concussion symptoms present at the time of testing. On the GSC, participants rate 22 symptoms on a scale from 0 to 6, with 0 indicating that the patient is not experiencing the listed symptom and 6 indicating the highest severity for that symptom. Thus, a higher score indicates worse concussion symptoms experienced by the participant.

After a brief explanation of the procedure, participants were asked to perform 2 STs: (1) Expanded Timed Get Up and Go (ETGUG) as a physical task and (2) APST as a cognitive task, twice. The order of STs was randomized. After completing the ST, participants were asked to combine ETGUG and APST to perform DT, twice. The order of tasks was standardized to control for the fatigue and practice effects between groups. Following DT performance, participants filled out the GSC again to reassess the number of symptoms and their severity. During DT, no instruction was provided to prioritize either motor or cognitive tasks.

Expanded timed get up and go (ETGUG)

Participants were instructed to sit with their backs against an armless chair with bare feet at the start point. With the verbal

cue, they stood up and started walking at their normal speed toward a cone placed 10 meters in front of the chair. Once they reached the cone, participants walked around it and returned to the chair, and sat back. The time of completion (from when a participant stood up to when they returned to their chair) was collected. The timer of the mobile phone was started with a verbal cue and stopped when the subject returned to the start point. The average completion time of two trials was used as an outcome measure.

Auditory pure switch task (APST)

Participants performed APST as a cognitive task. Participants were instructed to answer whether a randomly given number was odd or even. Four odd numbers (1, 3, 5, and 7) and four even numbers (2, 4, 6, and 8) were used for APST (Okumura et al., 2013). The new number was called out immediately after the previous response. Conventionally, the outcome measures used are response rate and response accuracy. Based on our pilot data, most participants had 100% response accuracy and minimal variability in response rate, therefore, we did not include this outcome in our analyses.

Dual-task (DT)

During DT, the participant was instructed to complete ETGUG while simultaneously answering APST. With the verbal cue, the participant stood up, walked, and returned to the chair as the investigator called the random numbers aloud throughout the ETGUG. The participant was not asked to prioritize ETGUG or APST during DT.

Statistical Analysis

All data were analyzed using Statistical Package for Social Science (SPSS) Version 28.0 (IBM®, Chicago, IL) with an alpha level of $p < 0.05$. Independent sample t-tests were conducted to analyze the differences in demographic characteristics between groups.

Paired sample t-tests were utilized to compare GSC before and after the data collection. Another paired sample t-test was conducted to compare pre-data collection GSC between 2 time points: subacute and after RTP. An independent sample t-test was conducted to compare the pre-data collection GSC between concussion and control groups.

To analyze the effects of concussion on ST and DT ETGUG performance, a mixed method (group x task) ANOVA was conducted. The analysis was conducted for each data collection session (subacute and after RTP). Another mixed method (group x time) ANOVA was conducted to examine the effect of concussion recovery time on ETGUG performance. The analysis was conducted for each task (ST and DT ETGUG). For the tests with significant *p*-value, the Turkey HSD post-hoc analysis was used.

Results

The average number of days for RTP data collection in the concussion group was 30.0 ± 12.8 days post-concussion. The interval between subacute and RTP data collection dates for the concussion group was 21.9 ± 12.4 days. The interval between 2 data collection dates for the control group was 19.4 ± 12.8 days, and there was no statistical difference in intervals between the two groups ($p = 0.605$).

Graded Symptom Checklist

Comparison of symptoms between concussion vs control groups

During the subacute phase, the concussion group experienced significantly higher total symptom scores ($p = 0.01$) and the number of symptoms ($p = 0.01$) compared to their matched control group (Figure 1(a)). However, at the RTP timepoint, there was no significant group difference (Figure 1(b)).

Comparison of symptoms before and after data collection

During the subacute phase, there was no significant difference in the number of symptoms before and after data collection in the concussion group ($p = 0.30$); however, total symptom scores decreased significantly after data collection compared to prior to data collection during the subacute phase ($p = 0.04$). After the completion of the RTP protocol, no significant differences in total symptom scores and the number of symptoms were found between before and after data collection. (Figure 2)

During the subacute data collection time point, the total symptom scores of the control group decreased slightly after data collection compared to before, but it did not reach statistical significance ($p = 0.07$). The number of symptoms significantly decreased after the data collection compared to before data collection ($p = 0.02$). At RTP data collection time point, no significant differences in total symptom scores and the number of symptoms were found between before and after data collection. (Figure 3)

ETGUG

The completion time of ETGUG during ST and DT in both concussion and control groups is presented in Table 2.

Task-by-group analysis

A mixed method ANOVA was performed to compare the effect of concussion on tasks at each time point (Table 5). There was a significant task difference ($p < 0.001$) and group difference ($p = 0.01$), but no significant task-by-group interaction ($p = 0.36$) at the subacute time point. At the completion of the RTP time point, there was a significant task difference ($p < 0.001$), group difference ($p = 0.001$), and task-by-group interaction effects ($p < 0.001$).

Post-hoc analysis on group effects revealed that the concussion group took significantly longer to complete ST and DT ETGUG compared to the control group at the subacute phase ($p = 0.02$ and $p = 0.01$, respectively), which is presented in Table 3. The concussion group took 3.49 ± 1.36 (95% CI 0.74–6.23) seconds longer to complete DT ETGUG than the control group. After the completion of the RTP protocol, the post-hoc analysis revealed that the concussion group took significantly longer to complete ST and DT ETGUG compared to the control group ($p = 0.01$ and $p < 0.001$, respectively). The concussion group took 5.65 ± 1.53 (95% CI 2.56–8.74) seconds longer to complete DT ETGUG than the control group.

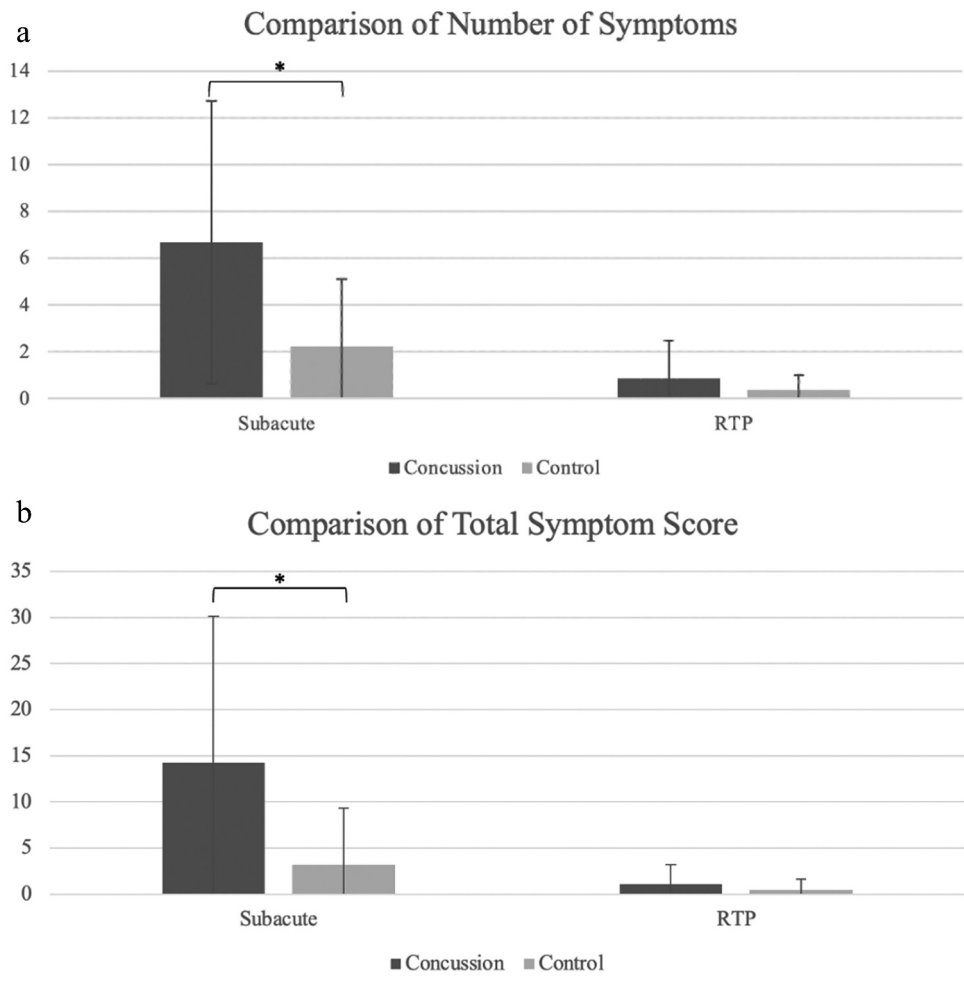

Figure 1. Comparison of GSC A) number of symptoms and B) total scores of symptoms. *Denotes significant values ($p < 0.05$). **GSC**: Graded Symptom Checklist, RTP: Return to Play.

The results from the post-hoc analysis on task effects are presented in Table 4. It revealed that the both concussion and control groups took significantly longer to complete DT compared to ST ETGUG at the subacute ($p = 0.048$ and $p = 0.04$, respectively). After the completion of the RTP protocol, only the concussion group showed a significant difference between ST and DT ETGUG performance ($p = 0.01$). This difference was not observed in the control group ($p = 0.07$).

Time-by-group analysis

A mixed method ANOVA was performed to compare the effects of concussion on time during ST and DT ETGUG (Table 2). The time-by-group analysis for ST ETGUG revealed that there was a significant group difference ($p = 0.01$) and time difference ($p = 0.002$), but no time-by-group interaction ($p = 0.79$). For DT ETGUG, there was a significant group difference ($p = 0.001$), but no time difference ($p = 0.11$), and time-by-group interaction ($p = 0.06$).

Post-hoc analysis on time effects revealed that the control group completed ST ETGUG significantly faster after RTP data collection time compared to the subacute data collection ($p = 0.003$), however, this difference was not observed in the concussion group ($p = 0.08$), which is presented in Table 5. A similar result was revealed for DT that the control group completed DT ETGUG significantly faster at the second data collection compared to the initial data collection ($p = 0.01$), but no significant time effects were observed in the concussion group ($p = 0.87$), indicating that ST and DT ETGUG have learning effects, but these effects might have been hindered by post-concussion impairments in individuals with concussion.

Discussion

The present study examined the effect of concussion on divided attention using clinically feasible DT among adolescent athletes with concussion and showed that the concussion group performed significantly worse compared to the control group. The current study further assessed the ST and DT performance after the completion of the RTP protocol and showed that post-concussion deficits still existed that were not identified with current concussion assessment tools. These residual deficits did not show in the GSC as there were no increases in symptoms before and after ST and DT performances, as well as between 2 time points.

The current study showed that the athletes with concussion took significantly longer to complete DT compared to the control group at both subacute and after the completion of the RTP protocol. Our results are consistent with previous DT

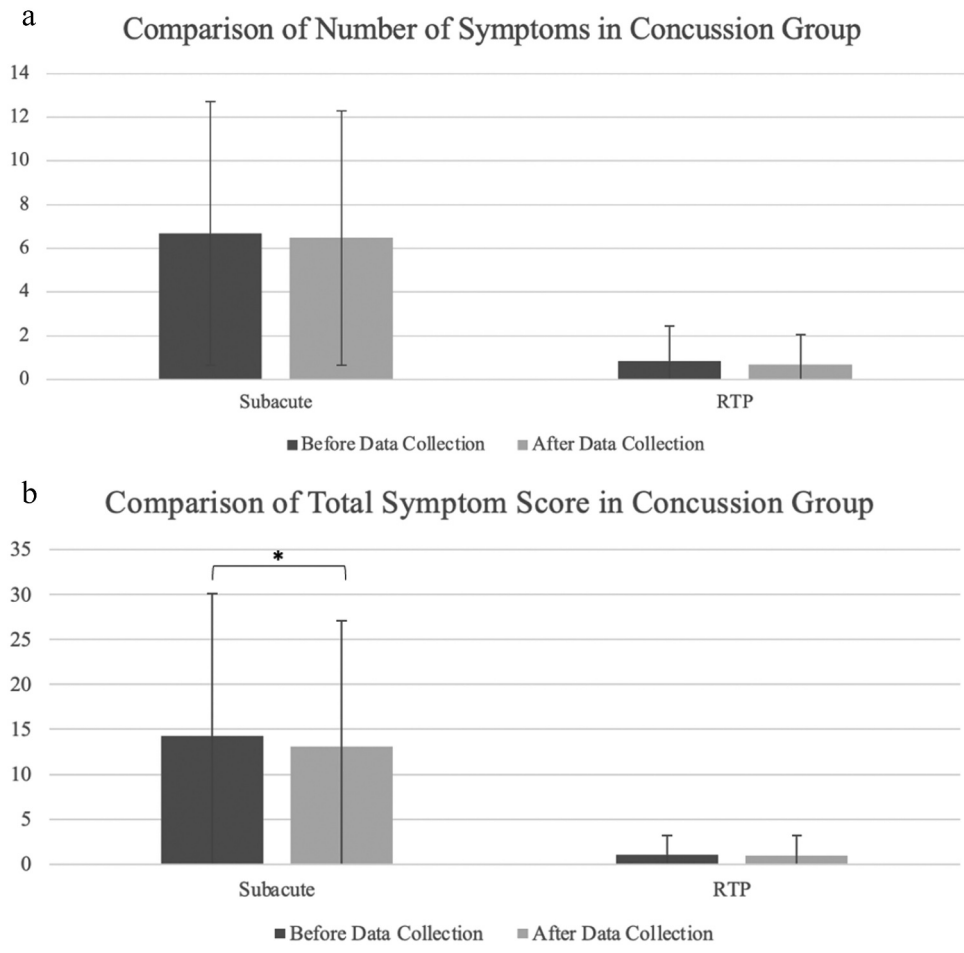

Figure 2. Comparison of GSC A) number of symptoms and B) total scores of symptoms in concussion group. *Denotes significant values ($p < 0.05$). **GSC**: Graded Symptom Checklist, RTP: Return to Play.

studies that used different dynamic postural control tools, including walking, tandem gait, and obstacle walking (Chiu et al., 2013; Hashida et al., 2021; Howell et al., 2019). While our results showed that the concussion group walked significantly slower during both ST and DT compared to the control group after completion of RTP, the deficits were more clearly shown for DT. It is reported that the divided attention deficits persist for up to 2 months, specifically for adolescents and younger adults (Howell et al., 2013). These studies indicate that divided attention impairments take longer to resolve, and persist even after the completion of the RTP protocol.

There was a significant difference in completion time between ST and DT in the concussion group, where they took approximately 5 seconds longer to complete DT compared to ST. While it is expected that cognitive or physical performance levels are compromised during DT condition, in the current study, the concussion group showed larger ST and DT deficits compared to the control group. Similar results were reported in a previous study that athletes with concussion performed DT gait worse than their matched non-concussed athletes and the group difference was not detected during ST (Catena et al., 2009). The underlying mechanism of how concussions affect DT performance is not completely understood; however, evidence suggests that performing DT leads to additional activation of the brain area (Blumen et al., 2014; Leone et al., 2017; Nijboer et al., 2014) to coordinate (Szameitat et al., 2002) and manage DT interference (Johannsen et al., 2013; Leone et al., 2017). Previous studies that used functional magnetic resonance imaging showed increased activity in the anterior cingulate, prefrontal, lateral frontal, dorsal premotor, and posterior parietal cortex during DT (Jiang, 2004; Jiang et al., 2004; Marois et al., 2006). These above-listed brain areas are activated only when DT is performed but not during ST. Since the current concussion assessment protocol relies on a battery of ST tests, the functions of these brain areas are not involved and consequently, it is possible that athletes may have returned to school or sport while still having post-concussion deficits associated with divided attention.

Given the increased stress on the brain function with DT, one concern we had was whether DT would exacerbate symptoms. Contrary to our concern, the concussion group had a significant decrease in total scores of symptoms, and the control groups had a significantly decreased number of symptoms, after data collection compared to before data collection. Previous studies also reported that subthreshold exercise could help to decrease symptoms and physiological dysfunction among different populations due to increased heart rate, cerebral blood flow improvement, increased

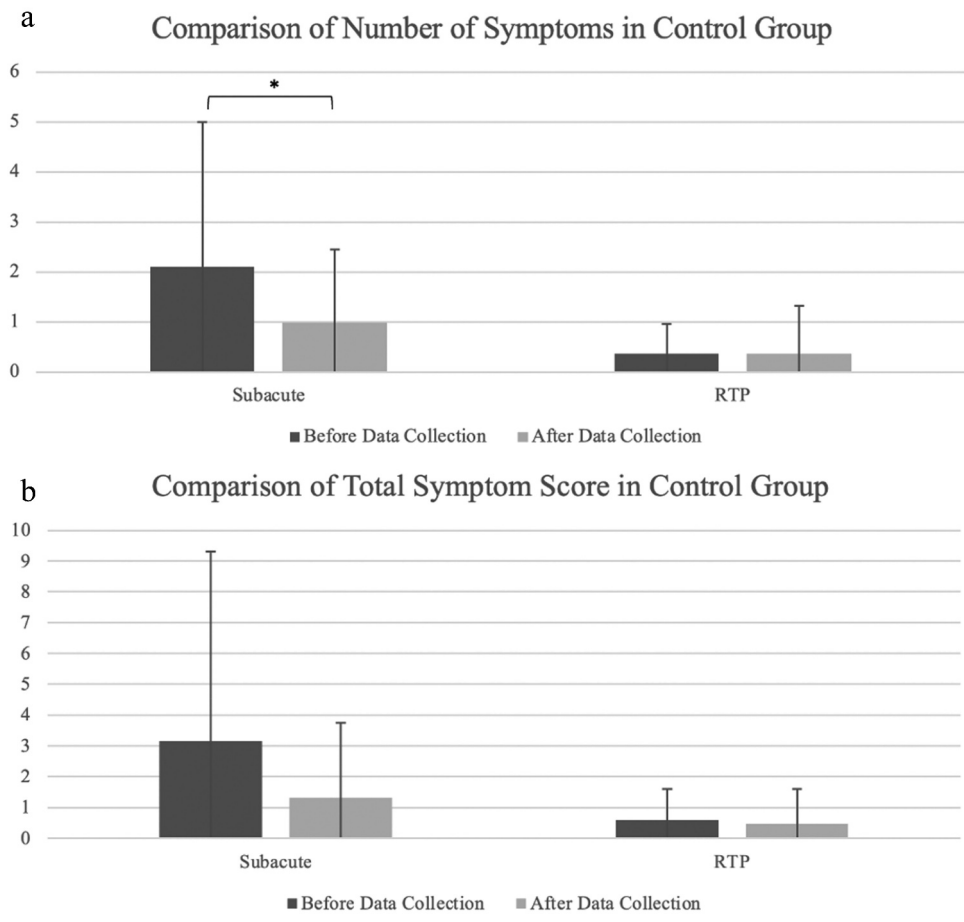

Figure 3. Comparison of GSC A) number of symptoms and B) total scores of symptoms in control group. **GSC**: Graded Symptom Checklist, RTP: Return to Play.

Table 2. Mean and SD of ST and DT ETGUG completion time measurement and ANOVA results between concussion and control groups at subacute and after RTP time points.

		Subacute	After RTP		Task Effects		
Group	Condition	(Mean ± SD)			Task	Group	Interaction
Control	ST	21.98 ± 2.46	20.32 ± 2.28	Subacute	<0.001*	0.010*	0.358
	DT	24.00 ± 3.55	21.98 ± 3.23	After RTP	<0.001*	0.001*	0.001*
					Time Effects		
					Time	Group	Interaction
Concussion	ST	24.59 ± 4.00	23.17 ± 3.90	ST	0.002*	0.005*	0.787
	DT	27.48 ± 4.92	27.63 ± 6.02	DT	0.106	0.001*	0.063

*Statistically significant effects were observed.
ST, Single-Task; DT, Dual-Task; ETGUG, Expanded Timed Get Up and Go;RTP, Return to Play.

Table 3. Post-hoc comparison of ST and DT ETGUG completion time between concussion and control groups at 2 time points.

		Mean Difference (Concussion-Control)	95% CI	*p*
Subacute	ST	2.61 ± 1.05	(0.48, 4.73)	0.018*
	DT	3.49 ± 1.36	(0.74, 6.23)	0.014*
After RTP	ST	2.85 ± 1.01	(0.79, 4.91)	0.007*
	DT	5.65 ± 1.53	(2.53, 8.78)	<0.001*

*The difference in completion time between concussion and control group was statistically significant.
ST, Single-Task; DT, Dual-Task; ETGUG, Expanded Timed Get Up and Go;RTP, Return to Play.

parasympathetic activity, and reduced sympathetic activity (Carter et al., 2003; Leddy et al., 2010). These results suggest that ETGUG combined with APST is safe to administer in individuals with concussion and possibly has positive effects on concussion recovery. Further studies should explore the use of DT in concussion rehabilitation.

Comparisons of symptoms between groups showed that the concussion group had higher total scores and the number of symptoms compared to the control group at the subacute time point, as expected. However, the group difference in symptoms was not observed after the completion of the RTP time point. While this indicates the decrease in symptoms in the concussion group to the 'normal' level, DT outcome still showed deficits compared to the control group. These results suggest that assessment of symptoms

Table 4. Post-hoc comparison of ETGUG completion time between DT and ST in each group at 2 time points.

		Mean Difference (DT-ST)	95% CI	*p*
Subacute	Concussion	2.90 ± 1.42	(0.22, 5.77)	0.048*
	Control	2.01 ± 0.97	(0.05, 3.97)	0.044*
After RTP	Concussion	4.46 ± 1.60	(1.22, 7.71)	0.008*
	Control	1.67 ± 0.88	(0.13, 3.45)	0.068

*The difference in completion time between ST and DT ETGUG was statistically significant.
ST, Single-Task; DT, Dual-Task; ETGUG, Expanded Timed Get Up and Go;RTP, Return to Play.

Table 5. Post-hoc comparison of ST and DT ETGUG completion time between subacute and after completion of RTP time points in concussion and control groups.

		Mean Difference (Subacute-After RTP)	95% CI	*p*
ST	Concussion	1.42 ± 3.39	(0.17, 3.01)	0.077
	Control	1.67 ± 2.23	(0.62, 2.71)	0.003*
DT	Concussion	1.15 ± 4.00	(−2.02, 1.72)	0.871
	Control	2.02 ± 3.10	(0.57, 3.47)	0.009*

*Control group improved ST and DT ETGUG performance significantly after the completion of RTP than at the subacute phase.
ST, Single-Task; DT, Dual-Task; ETGUG, Expanded Timed Get Up and Go;RTP, Return to Play.

in the later phase of concussion recovery may not provide useful information to detect residual post-concussion deficits.

This study has some limitations that need to be addressed. Our study did not control for the time interval between the 2 data collection sessions since each participant with concussion would have a different recovery trajectory. Furthermore, the timing of data collection after the completion of the RTP protocol fluctuated due to scheduling difficulties. While these individual timelines of athletes with concussion were matched with control participants, this might have affected the outcome of the second data collection in the concussion group. In addition, the use of multiple testing sites might have resulted in an inconsistent amount of distraction during the testing session. Finally, the sample size in the current study is rather small, and a future study with a larger sample size is warranted to address the effect of age (adolescents vs adults), sex, and concussion history on clinically feasible DT performance.

Conclusion

The clinically feasible DT test detected subtle and residual post-concussion deficits after athletes were cleared to RTP based on the guideline described in the 2017 Consensus Statement on Concussion in Sports. Divided attention is a necessary skill in daily and sports activities; thus, DT that assesses divided attention should be incorporated into concussion assessment protocol. The DT using ETGUG combined with APST are clinically feasible tools requiring minimal equipment and its simple outcome of completion time can differentiate athletes with concussion from non-concussed controls. Further study is warranted to examine when and if this divided attention deficit will resolve, whether baseline data is necessary for clinical decision making, and how this information should be used for concussion management.

Disclosure statement

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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